
Algorithm 1: Identification of Leading Provinces Using Cross-Correlation

Input:

- Compute and store incidence rate time series $\mathcal{Y} = \{y_1(t), y_2(t), \dots, y_n(t)\}$ for n provinces.
- Lag window $\mathcal{L} = \{-10, -9, \dots, 9, 10\}$.
- Correlation threshold $\tau = 0.7$.

Output:

- Number of lagging provinces N_i^- .
- Number of leading provinces N_i^+ .
- Leading province i^* .

for $i = 1$ **to** n **do** Define the reference time series $r(t) \leftarrow y_i(t)$. **for** $j = 1$ **to** n **do**

Compute the normalized cross-correlation

$$C_{ij}(\ell) = \text{xcorr}(y_j(t), r(t)), \quad \ell \in \mathcal{L}.$$

Determine the maximum correlation

$$C_{ij}^* = \max_{\ell \in \mathcal{L}} C_{ij}(\ell).$$

Determine the lag associated with the maximum correlation

$$\ell_{ij}^* = \arg \max_{\ell \in \mathcal{L}} C_{ij}(\ell).$$

Construct the filtered set

$$S_i = \{j : C_{ij}^* > \tau\}.$$

Compute the number of lagging provinces

$$N_i^- = |\{j \in S_i : \ell_{ij}^* < 0\}|.$$

if $N_i^- = 0$ **then**

Compute the number of leading provinces

$$N_i^+ = |\{j \in S_i : \ell_{ij}^* > 0\}|.$$

else

$$N_i^+ = 0.$$

Store

$$\text{Summary}_i = (N_i^-, N_i^+, S_i).$$

Determine the leading province

$$i^* = \arg \max_i N_i^+.$$

Return $\{\text{Summary}_i\}_{i=1}^n$ and i^* .
